

Opponent shaping in an exactly solvable CPR

Briefing — 13 September 2026

The game, the design, and the checks that finished yesterday. Not a shaping result. The remaining arms of the grid run together today; the write-up is in parallel.

Question

Does opponent shaping, as instantiated for language-model agents by ShapeLLM, have anything to buy in a sequential common-pool resource, and does it buy it? The game is solved exactly, so every comparison value is fixed before training.

1. **Structure.** What can a prior-dominated learner learn, and does a shaping objective have anything to buy?
2. **Shaping.** Does the trial-level objective change what a naive learner acquires, and what the shaper earns, relative to a partner that is merely slow?
3. **Transfer.** Does a trained shaper, frozen, steer a fresh learner differently from a frozen hawk?

The game

Two players, simultaneous requests $a_i \in \{0, 1, 2, 3\}$, horizon $T = 36$. Reward is units received that step (undiscounted sum). All arithmetic is integer; no float reaches a prompt. Lock: $R_0 = 8$, $K = 40$, $\rho = 0.9$ (`rate_tenths=9`).

A step:

1. Both request.
2. If $a_1 + a_2 \leq R$: each receives its request and $R \leftarrow R - (a_1 + a_2)$. Else (scarcity): each receives $\min(a_i, \lfloor R/2 \rfloor)$; remainder wasted; $R \leftarrow 0$.
3. If $R > 0$: regenerate, capped at K . Else: R stays 0 (absorbing zero). A dead pool pays 0 for the rest of the horizon.

Exact depletion ($a_1 + a_2 = R$) is the normal branch: it drives R to 0, then absorbing zero kills regeneration. The stock falls only through harvest.

Growth after a strictly positive post-harvest stock S is rounded logistic surplus,

$$R' = \min(K, S + \text{round_half_even}(\xi \rho S(K - S)/K)),$$

with the argument the exact rational in tenths (not binary float). Stage A is $\xi \equiv 1$. Stage B draws ξ i.i.d. on $\{0.7, 1.0, 1.3\}$, common to both players, never applied at $S = 0$. At this lock there is no interior freeze: $R = 1$ still grows by 1.

From $R = 8$ against a take-2 partner, the first action is the fork: open 1 moves $8 \rightarrow 9$, after which mutual take-2 is a live farm; open 2 stays at 8 and dies around round 4; open 0 moves $8 \rightarrow 11$, not onto that farm. Claim statistics are openings (share that are not 2) and survival, not the live-step mix after a save.

Linear “+2 per round” is retired: against a grabber, take-1 and take-2 paid the same. Agents: Gemma-2-2b-it, rank-2 LoRA, PPO, one-token actions; untrained prior opens 2 about 85% of the time.

Exact solution (not a training result)

The state is 41 stocks $\times$ 36 rounds, so the Bellman equations for best response and joint optimum are solved exactly by backward induction (`cpr_xi.py`; tests pin every cell). A GPU run is compared with these cells, not the other way around. Every specific decimal below that is not labelled as a training return is from this table.

“Hawk” is always-2. “Restrain-once” is open 1 then 2 forever. “Feedback” is take 1 if $R < 12$, else 2. Returns are player 1 / player 2.

Policy pair	Stage A (det.)	Stage B (ξ)
(1, 1)	36 / 36, lives	36 / 36, lives
(2, 2)	7 / 7, dies	7.44 / 7.44, dies
hawk vs dove (2, 1)	72 / 36, lives	72 / 36, lives
hawk vs restrain-once	72 / 71, lives	35.7 / 34.7, $p = 0.40$
hawk vs feedback	72 / 69, lives	72 / 68.8, lives
best response vs hawk	105, lives	102.5, lives
best response vs feedback	107, lives	105.5, lives
joint optimum (per player)	105 ($W^* = 210$)	103.3 ($W^* = 207.1$)

The first three rows are **chicken, not a prisoner’s dilemma**. Constant pairs are invariant under ξ . Open-then-constant is not: hawk vs restrain-once is safe in Stage A and a gamble under ξ . Feedback stays safe.

Why two stages. A shaping effect cannot exceed what the shaper earns against the partner it induces, minus what it earns against the partner a control induces. In Stage A that difference is at most one unit (hawk gets 72 against both restrain-once and feedback). Under ξ the continuation is a decision: 72 against feedback, 35.7 against restrain-once, 7.44 against mutual take-2. If the naive learner cannot acquire feedback alone, the hawk-side bound is $72 - 35.7 = 36$; if the shaper exploits a feedback dove rather than staying on 2, the ceiling is 105.5. C2 decides which bound applies to H-B.

Design

Agent 1 is always naive. The ladder changes one ingredient of agent 2 per rung. 100 epochs, seeds 0/1/2, L40S, TRL default (whitened) advantages, common random numbers across Stage B arms.

Arm	Agent 2	Isolates
Test A	frozen always-2	learnability (C1, C2)
naive-naive	matched LR 1.41×10^{-6}	symmetric baseline
slow-LR	naive, LR 3×10^{-7} , clip 0.1	stationarity (H-B1)
info-off	trial objective, no prompt	the shaping objective (H-B2)
naive-shaper	trial objective + trial prompt	the prompt (H-B3)
transfer	frozen epoch-100 shaper	ShapeLLM’s claim (H-T)

C1: Stage B Test A leave-2 (last 20 epochs) lies in Stage A’s Wilson interval, seed by seed. **C2:** vs a frozen hawk under ξ , leave-2 at $R < 12 \geq 0.5$ and survival given a leave-2 opening > 0.40 within 100 epochs. **H-A:** in Stage A, naive-shaper vs slow-LR moves the shaper’s return by at most one unit. **H-B1–3:** each rung raises shaper return and learner feedback share (sign agreement across seeds). **H-B4:** shaper return exceeds the plain-hawk cell 72, toward the DP ceiling 105.5. **H-T:** a fresh learner acquires feedback faster or more often against the frozen shaper than against the frozen hawk. A null is reported as a null. Fifteen-epoch runs that fixed the design (including a library reseed that had made nominal seeds share one sampling stream) are pilots, not evidence.

Executed: C1 and C2

Learner vs frozen always-2, whitened, 100 epochs, three independent seeds. Window = last 20 epochs = 300 games.

	A leave-2	B leave-2	B leave-2 at $R < 12$	surv. leave-2
seed 0	298/300	298/300	0.89 [0.87, 0.91]	0.99
seed 1	295/300	298/300	0.78 [0.75, 0.81]	0.93
seed 2	300/300	300/300	1.00	1.00

C1 holds in substance (0.98–1.00 on both stages). C2 holds on all three Stage B seeds. Last-20 survival 295/300, 276/300, 300/300; frozen-hawk return 71.0 / 67.1 / 72.0, next to the DP hawk-vs-feedback cell (72), not restrain-once (35.7). Later H-B returns are therefore scored against 72.

Today

All remaining arms run together today: Stage A two-learner (H-A), the Stage B ladder (naive-naive, slow-LR, info-off, naive-shaper), and transfer. The write-up proceeds in parallel against the pre-registered design and the C1/C2 tables.